

Assigned To: Junior SDET (You) Assigned By: QA Manager Project URL: https://demoqa.com Start Date: 2026-04-28
Deadline: 8 Weeks Priority: High

 WHY THIS WEBSITE?

Website Selected: DemoQA (<https://demoqa.com>) DemoQA is the industry's most widely used Selenium practice playground. It covers all real-world UI elements found in enterprise-grade applications — text boxes, dropdowns, alerts, date pickers, file upload/download, drag-and-drop, nested frames, and complex widgets — making it the ideal foundation for building a job-ready automation portfolio.

 TECHNOLOGY STACK

Java 17 Selenium WebDriver 4 TestNG Maven Page Object Model (POM) Git + GitHub Jenkins CI/CD Extent Reports Log4j

 PROJECT OBJECTIVE

Design, develop, and maintain a robust, end-to-end automation test suite for the DemoQA website using the Java-Selenium-TestNG stack. The framework must follow the **Page Object Model (POM)** design pattern, be version-controlled via **Git**, and be integrated into a **Jenkins CI/CD pipeline** for continuous test execution.

Upon completion, the candidate is expected to demonstrate a production-level automation framework that any QA team at a top-tier company could adopt.

 DELIVERY TIMELINE (8 WEEKS)

Week 1 Project Setup & Framework Skeleton
Week 2 Elements & Forms Module
Week 3 Alerts, Frames & Windows
Week 4 Widgets Module
Week 5 Interactions Module

Week 6 Books App Module
Week 7 Jenkins Pipeline + Reports
Week 8 Code Review & Demo

TEST CASE REPOSITORY — DEMOQA FULL SUITE

All test cases below must be automated. Each must follow the **Given / When / Then** structure in code comments and mapped to a unique Test Case ID.

TC ID	TEST CASE NAME	DESCRIPTION / STEPS	EXPECTED RESULT	PRIORITY
MODULE 1 — ELEMENTS				
TC_EL_001	Text Box Form Submission	Enter Full Name, Email, Current Address, Permanent Address and click Submit	Output section displays all entered values correctly	HIGH
TC_EL_002	Text Box — Invalid Email	Enter invalid email format (e.g., abc@) and submit	Email field is highlighted with an error border	MEDIUM
TC_EL_003	Check Box — Expand All & Select All	Click "Expand All" then select the root "Home" checkbox	All child checkboxes are selected; result section lists all items	HIGH
TC_EL_004	Check Box — Individual Node Selection	Expand tree and select only the "Downloads" folder	Only "Downloads" and its children appear in result	MEDIUM

TC ID	TEST CASE NAME	DESCRIPTION / STEPS	EXPECTED RESULT	PRIORITY
TC_EL_005	Radio Button — Yes Selection	Click the "Yes" radio button	"You have selected Yes" message is displayed	HIGH
TC_EL_006	Radio Button — Impressive Selection	Click the "Impressive" radio button	"You have selected Impressive" message is displayed	MEDIUM
TC_EL_007	Radio Button — No Option Disabled	Verify the "No" radio button is disabled	The "No" radio button is not clickable / is disabled	LOW
TC_EL_008	Web Table — Verify Data Rows	Fetch all rows from the initial web table and verify row count	Correct number of rows (3 by default) are present	HIGH
TC_EL_009	Web Table — Add New Record	Click Add, fill in First Name, Last Name, Age, Email, Salary, Dept, Submit	New record appears as a row in the table	HIGH
TC_EL_010	Web Table — Edit Existing Record	Click the Edit icon on Row 1, modify a field, save	Updated value reflects in the table row	HIGH
TC_EL_011	Web Table — Delete a Record	Click Delete icon on an existing row	Row is removed from the table	HIGH
TC_EL_012	Web Table — Search Filter	Type a name in the Search box	Table filters to show only matching rows	MEDIUM

TC ID	TEST CASE NAME	DESCRIPTION / STEPS	EXPECTED RESULT	PRIORITY
TC_EL_013	Buttons — Double Click	Double-click the "Double Click Me" button	"You have done a double click" message appears	HIGH
TC_EL_014	Buttons — Right Click	Right-click the "Right Click Me" button	"You have done a right click" message appears	HIGH
TC_EL_015	Buttons — Dynamic Click	Click the dynamically generated "Click Me" button	"You have done a dynamic click" message appears	MEDIUM
TC_EL_016	Links — Open Valid Link in New Tab	Click the "Home" link and switch to new tab	New tab opens with DemoQA home page	MEDIUM
TC_EL_017	Links — API Status: Created (201)	Click the "Created" API call link	Response status "Created" and code 201 shown	MEDIUM
TC_EL_018	Links — API Status: No Content (204)	Click the "No Content" API call link	Response status "No Content" and code 204 shown	LOW
TC_EL_019	Broken Images — Validate Broken Images	Load Broken Images page, get naturalWidth of all tags	At least one image has naturalWidth = 0 (broken)	MEDIUM
TC_EL_020	File Upload — Upload Valid File	Use sendKeys() to upload a .txt or .jpg file	Uploaded file name is displayed below the upload button	HIGH
TC_EL_021	File Download — Download File	Click the Download button and verify file lands in the Downloads directory	File is saved to local disk successfully	HIGH

TC ID	TEST CASE NAME	DESCRIPTION / STEPS	EXPECTED RESULT	PRIORITY
TC_EL_022	Dynamic Properties — Enable After 5s Button	Wait for the "Enable After 5 Seconds" button to become enabled	Button becomes clickable after 5 seconds (ExplicitWait)	HIGH
TC_EL_023	Dynamic Properties — Color Change Button	Capture the initial color of the button and verify it changes after 5s	Button color changes from initial to red	MEDIUM
TC_EL_024	Dynamic Properties — Visible After 5s	Wait for the "Visible After 5 Seconds" button to appear	Button becomes visible after 5 seconds	MEDIUM
MODULE 2 — FORMS				
TC_FM_001	Practice Form — Valid Full Submission	Fill First Name, Last Name, Email, Gender, Mobile, DOB, Subjects, Hobbies, Upload Photo, Address, State, City and Submit	Success modal "Thanks for submitting the form" appears with correct data	HIGH
TC_FM_002	Practice Form — Required Field Validation	Click Submit without filling any required fields	Required fields are highlighted in red	HIGH
TC_FM_003	Practice Form — Date of Birth Picker	Click on DOB field and select a specific date using the calendar	Selected date appears in the DOB input field	HIGH
TC_FM_004	Practice Form — Subject Auto-Complete	Type "Math" in subjects field and select from dropdown	"Maths" tag appears in the subjects field	MEDIUM
TC_FM_005	Practice Form — Multiple Hobbies Selection	Select all three hobbies: Sports, Reading, Music	All three checkboxes remain checked	MEDIUM

TC ID	TEST CASE NAME	DESCRIPTION / STEPS	EXPECTED RESULT	PRIORITY
TC_FM_006	Practice Form — State + City Dependent Dropdown	Select a State, then verify City dropdown is populated accordingly	City dropdown shows cities specific to the selected state	HIGH
TC_FM_007	Practice Form — Invalid Mobile (Less Than 10 Digits)	Enter 9-digit mobile number and submit	Mobile field is highlighted as invalid	MEDIUM
TC_FM_008	Practice Form — Modal Close Button	Submit valid form, then click Close on the success modal	Modal closes and user remains on the form page	LOW
TC_FM_009	Practice Form — Data Driven (Multiple Users)	Use TestNG @DataProvider to submit the form with 3 different datasets	Each dataset produces a successful submission modal	HIGH

MODULE 3 — ALERTS, FRAME & WINDOWS

TC_AF_001	Browser Alert — Accept Simple Alert	Click "Click me" to trigger an alert and accept it	Alert is dismissed and success message appears	HIGH
TC_AF_002	Browser Alert — Timer Alert	Click the 5-second timer alert button and accept after it appears	Alert accepted successfully after delay	MEDIUM
TC_AF_003	Confirm Alert — Accept	Click the confirm alert button and accept	Result text shows "You selected Ok"	HIGH
TC_AF_004	Confirm Alert — Dismiss	Click the confirm alert button and dismiss	Result text shows "You selected Cancel"	HIGH

TC ID	TEST CASE NAME	DESCRIPTION / STEPS	EXPECTED RESULT	PRIORITY
TC_AF_005	Prompt Alert — Enter Text & Accept	Trigger prompt, type "DemoQA Tester", accept	Result text shows "You entered DemoQA Tester"	HIGH
TC_AF_006	New Tab — Switch and Verify	Click "New Tab" button, switch to new tab	New tab opens; title reads "This is a sample page"	HIGH
TC_AF_007	New Window — Switch and Verify	Click "New Window" button, switch to window handle	New window opens with correct page content	HIGH
TC_AF_008	New Window Message — Verify Content	Click "New Window Message" and read the content	Message content is displayed correctly in new window	MEDIUM
TC_AF_009	Frames — Interact With Frame 1	Switch into Frame 1, read the heading text	Heading text inside frame is captured correctly	HIGH
TC_AF_010	Frames — Interact With Frame 2	Switch into Frame 2, read the heading text	Heading text inside frame 2 is captured correctly	HIGH
TC_AF_011	Nested Frames — Parent and Child	Switch to parent frame, then switch to child frame and read text	Child frame text "Child Iframe" is captured correctly	HIGH
TC_AF_012	Modal Dialog — Small Modal Open & Close	Click "Small Modal" button, verify it opens, then close it	Modal opens with correct title; closes on clicking Close	MEDIUM

TC ID	TEST CASE NAME	DESCRIPTION / STEPS	EXPECTED RESULT	PRIORITY
TC_AF_013	Modal Dialog — Large Modal Open & Close	Click "Large Modal" button, read body content, then close it	Large modal opens with lorem ipsum text; closes correctly	MEDIUM
MODULE 4 — WIDGETS				
TC_WG_001	Accordion — Expand Section 1	Click on Section 1 heading	Section 1 content is revealed	MEDIUM
TC_WG_002	Accordion — Expand/Collapse Toggle	Click Section 2 (collapsed), verify content; click again, verify collapsed	Accordion correctly toggles open and closed	MEDIUM
TC_WG_003	Auto Complete — Multi Value	Type and select multiple color names in the multi-value input	Each color tag appears correctly in the input	HIGH
TC_WG_004	Auto Complete — Single Value	Type "Red" in the single-value input, select from dropdown	"Red" is selected in the single-value field	HIGH
TC_WG_005	Date Picker — Select Date	Click date input and pick a specific date from the calendar	Selected date populates the date input field	HIGH
TC_WG_006	Date & Time Picker — Select Date and Time	Click Date-Time input, select a date and a time	Date and time appear in the input field	HIGH
TC_WG_007	Slider — Move to Specific Value	Use Actions to drag the slider to value 75	Input box displays value 75	HIGH

TC ID	TEST CASE NAME	DESCRIPTION / STEPS	EXPECTED RESULT	PRIORITY
TC_WG_008	Progress Bar — Start and Stop	Click Start, wait until progress reaches 50%, click Stop	Progress bar stops at or near 50%	MEDIUM
TC_WG_009	Progress Bar — Reset	Complete progress to 100%, then click Reset	Progress bar resets to 0% and Start button reappears	MEDIUM
TC_WG_010	Tabs — What Tab Active	Verify the "What" tab is active on page load, read its content	Tab content for "What" is visible and correct	MEDIUM
TC_WG_011	Tabs — Origin Tab Navigation	Click the "Origin" tab	"Origin" tab content becomes visible	MEDIUM
TC_WG_012	Tool Tips — Button Hover	Hover over the "Hover me to see" button and capture tooltip text	Tooltip "You hovered over the Button" is displayed	HIGH
TC_WG_013	Tool Tips — Input Field Hover	Hover over the text field and capture tooltip	Tooltip text is displayed correctly	MEDIUM
TC_WG_014	Menu — Hover Over Main Item 2	Hover over "Main Item 2" to reveal sub-menu	Sub-menu items are visible	MEDIUM
TC_WG_015	Select Menu — Select Option by Value	Open "Select Value" dropdown and pick "Dr."	"Dr." is selected and displayed	MEDIUM
TC_WG_016	Select Menu — Multi Select	Select multiple values from the multi-select dropdown	All selected values appear as tags	MEDIUM

TC ID	TEST CASE NAME	DESCRIPTION / STEPS	EXPECTED RESULT	PRIORITY
MODULE 5 — INTERACTIONS				
TC_IN_001	Sortable — Drag List Item to Top	Drag item "Cras justo odio" to the first position	Item moves to position 1 in the list	HIGH
TC_IN_002	Sortable — Grid View Sort	Switch to Grid tab, drag one tile to another position	Tile is repositioned correctly in the grid	MEDIUM
TC_IN_003	Selectable — Select Single Item	Click on item "Cras justo odio" in the list	Item becomes highlighted / active	MEDIUM
TC_IN_004	Selectable — Select Multiple Items	Click multiple list items holding Ctrl	Multiple items are highlighted simultaneously	MEDIUM
TC_IN_005	Resizable — Resize Box 1	Drag the resize handle of the first box to make it larger	Box dimensions increase accordingly	MEDIUM
TC_IN_006	Droppable — Simple Drag and Drop	Drag the "Drag me" element and drop onto "Drop here" target	Target text changes to "Dropped!"	HIGH
TC_IN_007	Droppable — Accept Only Acceptable Draggable	Switch to Accept tab, drag acceptable item to drop zone	Drop zone shows "Dropped!" for acceptable item only	HIGH
TC_IN_008	Draggable — Move Element by Offset	Use Actions.dragAndDropBy() to move the draggable element by (100, 100)	Element position changes by the given offset	HIGH

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TC_IN_009	Draggable — Axis Restricted (X Only)	Try to drag an X-axis-restricted element vertically	Element only moves horizontally	LOW
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MODULE 6 — BOOK STORE APPLICATION

TC_BS_001	Login — Valid Credentials	Enter valid username and password (demoqa/pass), click Login	User is logged in and profile page is shown	HIGH
TC_BS_002	Login — Invalid Credentials	Enter invalid credentials and click Login	Error message "Invalid username or password!" is displayed	HIGH
TC_BS_003	Login — Empty Fields Submission	Leave username and password blank and click Login	Fields are highlighted with validation errors	MEDIUM
TC_BS_004	Book Store — Verify Books Are Listed	Navigate to Book Store, check total book count	At least 8 books are listed on the page	HIGH
TC_BS_005	Book Store — Search for Book	Type "Git" in the search bar	Only books matching "Git" are displayed	HIGH
TC_BS_006	Book Store — Click Book and Verify Details	Click on "Git Pocket Guide", verify title, ISBN, publisher on detail page	Book detail page shows correct metadata	HIGH
TC_BS_007	Profile — Add Book to Collection	Login, navigate to a book, click "Add To Your Collection"	Book appears in user's Profile collection	HIGH

TC ID	TEST CASE NAME	DESCRIPTION / STEPS	EXPECTED RESULT	PRIORITY
TC_BS_008	Profile — Delete Book From Collection	Login, go to Profile, click Delete icon on an added book, confirm	Book is removed from the collection	HIGH
TC_BS_009	Profile — Delete All Books	Click "Delete All Books" button and confirm the dialog	All books removed from the collection; table shows empty state	MEDIUM
TC_BS_010	Logout	Click Logout button from the Profile page	User is redirected to Login page	HIGH
TC_BS_011	Book Store — Rows Per Page	Change "rows per page" dropdown to 20	Up to 20 book rows are displayed per page	LOW
TC_BS_012	Book Store — Pagination Navigation	Click Next page button on Book Store	Next page of books is loaded	MEDIUM

MODULE 7 — CI/CD & FRAMEWORK REQUIREMENTS

TC_CI_001	Jenkins Pipeline — Trigger Build on Git Push	Push code to main branch; Jenkins pipeline auto-triggers	Jenkins build starts automatically via webhook	HIGH
TC_CI_002	Jenkins — Maven Test Execution	Run `mvn test` from Jenkinsfile	All tests are executed and results published	HIGH
TC_CI_003	TestNG XML — Parallel Execution	Configure testng.xml with parallel="methods" threads="3"	Tests run in parallel across 3 threads	HIGH

TC ID	TEST CASE NAME	DESCRIPTION / STEPS	EXPECTED RESULT	PRIORITY
TC_CI_004	Extent Report — HTML Report Generated	Run full suite; verify Extent report HTML file is generated	HTML report with pass/fail stats is created in /reports folder	HIGH
TC_CI_005	Screenshot on Failure	Intentionally fail a test; verify screenshot is saved	Screenshot .png captured in /screenshots folder with test name	HIGH
TC_CI_006	Cross Browser — Chrome & Edge	Run the smoke suite on both Chrome and Edge via testng.xml parameters	All smoke tests pass on both browsers	HIGH

☒ DELIVERABLES EXPECTED

60+

Automated Test Cases

POM

Page Object Model Framework

Git

Repository with meaningful commit history

Jenkins

CI/CD Pipeline with Jenkinsfile

TestNG

testng.xml with grouping & parallel execution

Reports

Extent HTML Report + Log4j Logs

EXPECTED FRAMEWORK FOLDER STRUCTURE

```
demoqa-automation/
├── src/
│   ├── main/java/
│   │   ├── base/           ← BaseTest.java (driver setup, teardown)
│   │   ├── pages/          ← One Page class per module (TextBoxPage, FormPage...)
│   │   └── utils/           ← ConfigReader, WaitHelper, ScreenshotUtil, ExcelUtil
│   └── test/java/
│       └── tests/           ← One Test class per module (ElementsTest, FormsTest...)
├── resources/
│   ├── config.properties ← URL, browser, timeout settings
│   └── testdata/          ← Excel/JSON test data files
├── testng.xml             ← Suite config with groups & parallel settings
├── Jenkinsfile            ← Pipeline stages: Checkout, Build, Test, Report
├── pom.xml                ← Maven dependencies (Selenium, TestNG, Extent, Log4j)
└── reports/               ← Auto-generated Extent HTML reports
```

 **Manager Note** Commit your code to GitHub daily. Use feature branches (e.g., feature/elements-module) and open Pull Requests for each module. All code must be reviewed before merging to main. This project will be part of your portfolio for SDET interviews — treat it like production code.